Mantena Monish

B.Tech - CSE(DS)

Contact

+919440332123

Mantena.Monish_2026@woxsen.edu.in

Hyderabad, Telangana| India

www.linkedin.com/in/monish-mantena

https://github.com/MantenaMonish

Academics

Woxsen University | Hyderabad

2022 - 2026 |**Bachelor of Technology - 3.6/4.0**

FIITJEE Junior College | Hyderabad

2020 - 2022 |Intermediate **- 91%**

Nasr Boys School | Hyderabad

2010 - 2020 | Schooling **- 87%**

Skills

- Programming Languages: Python, .NET
- Data Analysis & Visualization
- Data Ingestion & Transformation
- Data Modeling & Warehousing
- Machine Learning & Deep Learning
- Database Management
- Neural Networks
- Data Structures & Algorithms
- Computer Vision(CV)

Tools

- Machine Learning: Scikit-Learn, TensorFlow
- Data Engineering: Pandas, NumPy
- Databases: MySQL
- Visualization: Tableau, Excel
- CI & CD: Git, Docker
- Notebooks: Jupyter Notebook, Google Colab

Languages

- English - Fluent
- Telugu - Fluent
- Hindi - Fluent

Professional Skills

- Flexible Planning and Coordination
- Teamwork and Collaboration
- Problem Solving
- Critical Thinking

Objective

Fourth-year B.Tech student in Computer Science and Engineering (Data Science specialization) with strong skills in Computer Vision, Deep Learning, and Python. Experienced in training CV models using PyTorch, TensorFlow, YOLOv8, and OpenCV, along with data preparation, annotation, and evaluation. Proficient in Git and ML workflows, with hands-on work in object detection, tracking, and real-time video analysis. Eager to contribute to AI/CV model development, optimization, and integration into real-world applications.

Projects

Project - Real-Time Vehicle Detection & Tracking System

- Optimized the end-to-end real-time object detection pipeline using YOLOv8 and DeepSORT by tuning detection confidence thresholds, NMS parameters, and tracker matching metrics—achieving 30 FPS throughput with stable multi-object tracking.
- Improved tracking robustness using DeepSORT's motion prediction (Kalman Filter) and appearance embeddings, reducing ID switching to under 5% and achieving 92% IDF1 on UA-DETRAC test data.
- Implemented a calibrated vehicle speed estimation module using pixel displacement and timestamp deltas, applying camera perspective correction to achieve ±3 km/h accuracy in speed prediction.
- Conducted extensive testing across diverse lighting conditions and occlusions, identified bottlenecks, and performed debugging and threshold tuning to significantly enhance inference reliability.

Project - From Raw Data to Accurate Forecasts: A Framework for Temperature Prediction Using Advanced Modeling Techniques

- Forecasted outdoor temperatures at Woxsen using historical weather station data (Feb 2024–Aug 2024) from the university, applying advanced modeling techniques.
- Compared the accuracy of 5 machine learning algorithms, including LSTM, Bidirectional LSTM, and Auto ARIMA models, evaluating their performance using metrics like MAPE, RMSE, and R^2.
- Achieved highly accurate predictions with the LSTM model, obtaining 12.33% MAPE, 0.0907 RMSE, and 0.9570 R^2, demonstrating strong model efficiency and reliability.

Project - Enterprise Banking Web API System

- Designed and developed a layered banking Web API using ASP.NET 8 implementing Admin, User, Account, Fixed Deposit, and Fund Transfer modules.
- Implemented transaction-safe operations for fund transfers and FD creation using SQL Server stored procedures with commit/rollback mechanisms.
- Built a DB-First architecture with ADO.NET repositories, centralized connection management, and DTO-based API responses ensuring scalability and separation of concerns.

Research & Publications

- Mantena Monish, et al. "A Comparative Analysis of Machine Learning Models for High-Frequency Temperature Forecasting" (Manuscript prepared for journal submission).

Certifications

- Data Science for Engineers - NPTEL
- Data to Dashboard: Tableau - NPTEL
- Data Structures - Coursera
- Machine Learning with Python - Coursera
- Introduction to NoSQL Databases - Coursera

Achievements

- **MGIT Hackathon Finalist** – Developed innovative solutions in a competitive coding environment.
- **Finalist, Talent & Development Challenge (Woxsen University, 2025)** – Ranked among the top 80 out of 1,300+ participants.
- **Academic Excellence Awardee** – Recognized as a top performer in the B.Tech program.